ST-LINK v2 Debugger, SWD Protocol & Embedded In-Circuit Emulation

1. Overview of ST-LINK v2 & In-Circuit Debugging

The **ST-LINK v2** is an in-circuit debugger and programmer for the STM8 and STM32 microcontroller families. Rather than running a traditional software monitor or bootloader on the target, the ST-LINK communicates directly with hardware debugging silicon integrated on the ARM Cortex-M3 die through the **ARM CoreSight Serial Wire Debug (SWD)** interface.

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2. SWD Physical Layer & Protocol Specification

```
         HOST PC (STM32CubeIDE)
                  | (USB 2.0 Full-Speed 12 Mbps)
             [ST-LINK v2]
                  |
     +------------+------------+
     |                         |
   SWDIO (PA13)              SWCLK (PA14)
(Bidirectional Data)        (Clock: 1-4 MHz)
```

2.1 Comparison: SWD vs. JTAG (IEEE 1149.1)

Parameter	SWD (Serial Wire Debug)	JTAG (Joint Test Action Group)
Dedicated Pins	**2 pins** (`SWDIO`, `SWCLK`)	**4 to 5 pins** (`TDI`, `TDO`, `TCK`, `TMS`, `nTRST`)
Pins Saved for Application	**Frees 3 MCU pins** for GPIOs/Peripherals	Consumes valuable I/O pins
Maximum Data Rate	Up to 4.0 MHz clock frequency	Typically 1–10 MHz
Boundary Scan Testing	Not supported	Supported (ASIC pin-level PCB testing)
Protocol Framing	Packet-based with parity & ACK	Shift-register based state machine (TAP)
Silicon Implementation	ARM CoreSight proprietary DP/AP	Standard IEEE 1149.1 Test Access Port

2.2 SWD Packet Structure & Electrical Turnaround

Every SWD transaction is synchronous to **`SWCLK`** and consists of three phases:

1. **8-bit Packet Request (Host $\rightarrow$ Target)**:

- Start bit (1)
- APnDP bit (0 = Debug Port, 1 = Access Port)
- RnW bit (0 = Write, 1 = Read)
- 2-bit Address (**`A[3:2]`**)
- Even Parity bit
- Stop bit (0) + Park bit (1)

2. **Turnaround (`Trn`) Period**: 1 clock cycle where neither device drives the line.

3. **3-bit ACK Response (Target $\rightarrow$ Host)**:

- `001` = **OK** (Success)
- `010` = **WAIT** (Target busy)
- `100` = **FAULT** (Transfer error)

4. **32-bit Data Phase + Parity**: The payload transmitted with LSB first.

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3. Physical Wiring to the STM32 Blue Pill

ST-LINK Pin	Blue Pill Pin	Description	Critical Precautions
SWDIO	`PA13`	Bidirectional Data	Keep wire length under 15 cm to prevent signal reflection
SWCLK	`PA14`	Synchronous Clock	Keep clear of fast-switching PWM or buzzer lines
GND	`GND`	Ground Reference	**MANDATORY**: Common reference prevents logic shift
3.3V	`3.3V`	Target Supply	Power target from ST-LINK OR USB (Never both!)

> *[!CAUTION]*
>
> ***Ground Loop & Dual-Power Hazard:*** *Connecting the ST-LINK 3.3V supply and the Blue Pill USB 5V cable simultaneously can back-feed voltage regulators, causing reverse current flow and thermal breakdown. For live debugging with USB connected, disconnect the ST-LINK 3.3V pin and leave only* ***SWDIO, SWCLK, and GND****.*

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4. Boot Selection Modes (`BOOT0` and `BOOT1`)

The STM32F103 boot configuration is evaluated on the fourth rising edge of `SYSCLK` after power-on reset:

BOOT0 Jumper	BOOT1 Jumper	Boot Space	Starting Address	Target Application
0 (Default)	**X**	**Main Flash Memory**	`0x08000000`	Normal execution of compiled user firmware
1	**0**	**System Memory**	`0x1FFFF000`	Factory ST ROM bootloader (UART flashing via `PA9`/`PA10`)
1	**1**	**Embedded SRAM**	`0x20000000`	Direct RAM execution (diagnostic code development)

- **Standard Project Setting:** Keep both `BOOT0 = 0` and `BOOT1 = 0`.

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5. ARM Cortex-M3 On-Chip Hardware Debug Silicon

Unlike software debuggers that modify assembly instructions in RAM, the Cortex-M3 includes hardware debugging macrocells:

1. **FPB (Flash Patch and Breakpoint Unit)**:

- Provides **6 hardware instruction comparators**.

• Compares the Program Counter (PC) directly on the instruction bus. When a match occurs, it halts the CPU pipeline without modifying Flash memory.

2. **DWT (Data Watchpoint and Trace Unit)**:

• Provides **4 data watchpoints**.

• Monitors memory addresses (e.g. `&currentRtc`) and triggers a halt or trace packet when a variable is written or read.

3. **NVIC Vector Catch**:

• Automatically halts the processor if a `HardFault`, `BusFault`, `UsageFault`, or `MemManage` exception occurs, displaying the exact faulted instruction address in STM32CubeIDE.

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6. Evaluator Oral Questions & Detailed Answers

Q1: What is the difference between a hardware breakpoint and a software breakpoint?

> ***Answer**:*
>
> *A **software breakpoint** works in RAM by replacing the target instruction with the ARM `BKPT` opcode (`0xBE00`). When the CPU reaches this opcode, it traps into debug mode. This cannot be done in Flash memory without an erase-write cycle. A **hardware breakpoint** uses the Cortex-M3 **FPB unit**, where dedicated hardware address comparators monitor the bus and trigger a core halt when the Program Counter equals the breakpoint address, leaving Flash memory completely unmodified.*

Q2: What should you do if the debugger returns "Cannot connect to target"?

> ***Answer**:*
>
> *Common causes and engineering remedies:*
>
> *1. **Floating Ground**: Ensure ST-LINK GND is firmly seated to the Blue Pill GND rail.*
>
> *2. **Disabled SWD Pins**: If code remaps `PA13`/`PA14` as standard GPIOs, the SWD port is locked out after boot.*
>
> *3. **Deep Sleep**: If the MCU enters `STOP` or `STANDBY` mode, clocks to the debug port are gated off.*
>
> *4. **Fix (Connect Under Reset)**: Hold the Blue Pill onboard `RESET` button, initiate debug launch in STM32CubeIDE set to "Software System Reset / Connect Under Reset", and release the button once the ST-LINK establishes the hardware handshake.*

Q3: How does the debugger inspect variables in real time without pausing code execution?

> ***Answer**:*
>
> *Through the **CoreSight Memory Access Port (MEM-AP)**. The MEM-AP has direct access to the internal AHB system bus, allowing it to read and write SRAM locations (like `currentRtc` or `currentState`) during idle bus cycles without stealing clock cycles from the Cortex-M3 CPU core (Live Expressions).*